

Chapter 9: The Behavior of Solutions Part III

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9.9. Regular Solutions

- ✓ Up to this point, two classes of solutions have been identified. The first is the **ideal solution**,

$$a_i = X_i \quad \Delta \bar{H}_i^M = 0 \quad \Delta \bar{V}_i^M = 0 \quad \Delta \bar{S}_i^M = -R \ln X_i$$

for all X_i , all temperatures and all pressures.

- ✓ All other solutions can be classified as **nonideal solutions**, in which the relationships $a_i = X_i$ and $\Delta \bar{H}_i^M = 0$ are **not valid**.
- ✓ Attempts to classify nonideal solutions have involved the development of equations that describe the behavior of hypothetical solutions, and the simplest of these mathematical formalisms is that which generates what is known as **regular solution behavior**.

9.9. Regular Solutions

- ✓ A regular solution is one in which

$$\Delta H^M = \alpha X_A X_B$$

$$\Delta S^M = \Delta S^{M,ideal}$$

where α is a constant.

- ✓ Thus, for a **regular solution**:

$$\Delta G^M = \alpha X_A X_B - T \Delta S^{M,ideal}$$

- ✓ It is common to define excess functions as the value of the state function minus that of the ideal value of the state function.

$$G^{XS} = G - G^{id}$$

9.9. Regular Solutions

✓ For a regular solution, the excess molar Gibbs free energy of mixing is

$$G^{XS} = \Delta G^M - \Delta G^{M,ideal} = \Delta H^M = \alpha X_A X_B$$

For a regular solution,

$$\frac{\partial G^{XS}}{\partial T} = -S^{XS} = 0$$

We see that, at any composition,

$$\overline{G}_A^{XS} = RT \ln \gamma_A$$

is independent of temperature. Thus,

$$\overline{G}_A^{XS} = RT_1 \ln \gamma_A(T_1) = RT_2 \ln \gamma_A(T_2)$$

from which we see that

$$\frac{\ln \gamma_A(T_2)}{\ln \gamma_A(T_1)} = \frac{T_1}{T_2}$$

9.9. Regular Solutions

$$\frac{\ln \gamma_A(T_2)}{\ln \gamma_A(T_1)} = \frac{T_1}{T_2}$$

- ✓ This equation is of considerable practical use in **converting activity data** for a regular solution at **one temperature** to activity data at **another temperature**.
- ✓ We also see that as T_2 becomes very large, $\gamma_A(T_2)$ approaches unity.

$$\Delta \bar{H}_B = \Delta H^M + (1 - X_B) \frac{d\Delta H^M}{dX_B}$$

- ✓ Since $\Delta H^M = \alpha X_A X_B$, this yields $\Delta \bar{H}_B = \alpha X_A^2$

9.9. Regular Solutions

- ✓ Since all the state functions of a regular solution are symmetric with respect to the interchanging of X_A and X_B , we can also write

$$\Delta \bar{H}_A = \alpha X_B^2$$

- ✓ Now,

$$\Delta \bar{S}_i^M = -R \ln X_i$$

$$\Delta \bar{G}_A^M = RT \ln a_A = \Delta \bar{H}_A^M - T \Delta \bar{S}_A^M$$

$$RT \ln a_A + T \Delta \bar{S}_A^M = \Delta \bar{H}_A^M$$


$$RT \ln \gamma_A + RT \ln X_A - RT \ln X_A = \Delta \bar{H}_A^M$$

$$RT \ln \gamma_A = \Delta \bar{H}_A^M = \alpha X_B^2$$

9.9. Regular Solutions

✓ Finally, this give us $RT \ln \gamma_A = \alpha X_B^2$

$$\gamma_A = \exp\left(\frac{\alpha X_B^2}{RT}\right) \qquad \gamma_B = \exp\left(\frac{\alpha X_A^2}{RT}\right)$$

✓ The following relationships are valid for a **regular solution**.

As $X_A \rightarrow 1$, $\gamma_A = \exp\left(\frac{\alpha \cdot 0}{RT}\right) \rightarrow 1$, A is a Raoultian solution

As $X_A \rightarrow 0$, $\gamma_A = \exp\left(\frac{\alpha \cdot 1}{RT}\right) \rightarrow a$ constant, A is a Henrian solution

As $X_B \rightarrow 1$, $\gamma_B = \exp\left(\frac{\alpha \cdot 0}{RT}\right) \rightarrow 1$, B is a Raoultian solution

As $X_B \rightarrow 0$, $\gamma_B = \exp\left(\frac{\alpha \cdot 1}{RT}\right) \rightarrow a$ constant, B is a Henrian solution

9.9. Regular Solutions

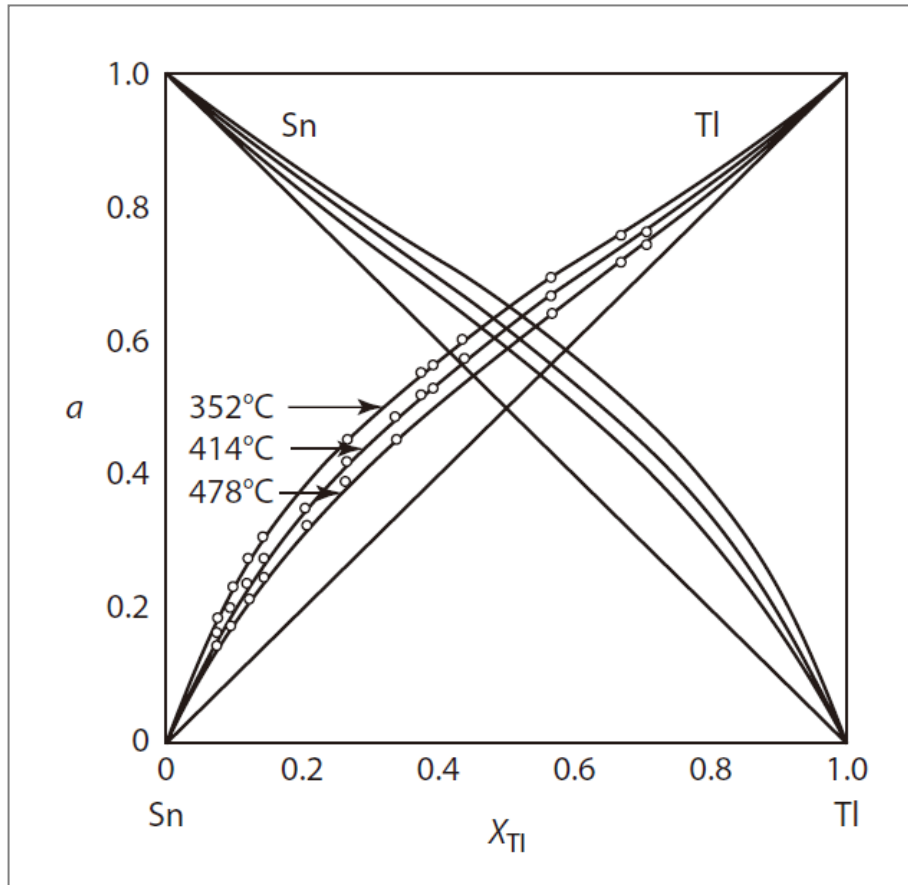


Figure 9.13
Activities in the liquid system tin–thallium.
As T increases, a towards unity

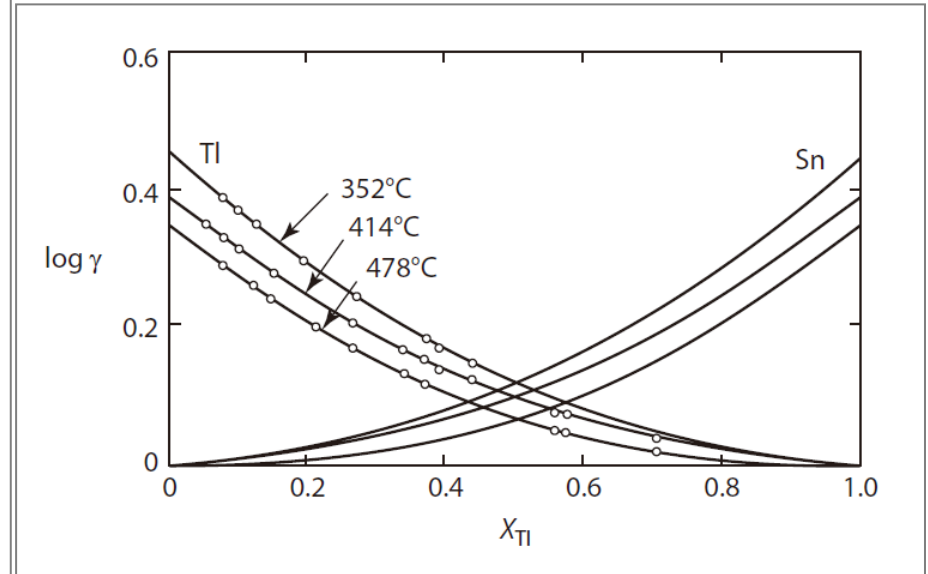
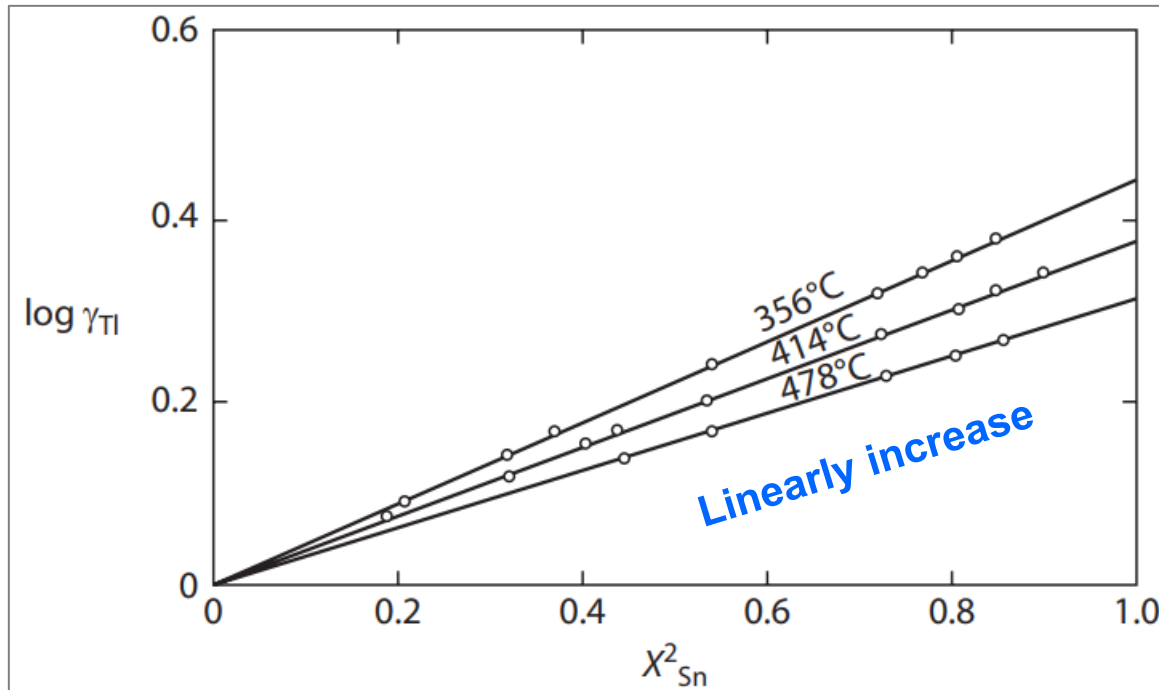


Figure 9.14
Activity coefficients in the liquid system tin–thallium.

9.9. Regular Solutions

- ✓ This Figure shows the linear variations of $\log \gamma_{Tl}$ with X_{Sn}^2 , the slopes of which equal α at the given temperatures.

Slope α is the constant



$$RT \log \gamma_A = \alpha X_B^2$$

$$\log \gamma_{Tl} = \alpha_1 X_{Sn}^2 / RT$$

$$\log \gamma_{Tl} = \alpha_2 X_{Sn}^2 / RT$$

$$\alpha = \alpha_2 / RT$$

Figure 9.15 $\log \gamma_{Tl}$ vs. X_{Sn}^2 in the system tin–thallium. (From J. H. Hildebrand and J. N. Sharma, “The Activities of Molten Alloys of Thallium with Tin and Lead,” *J. Am. Chem. Soc.* (1929), vol. 51, p. 462.)

9.9. Regular Solutions

- ✓ The variation of y_i with X_i is that of a regular solution, but Figure 9.16 shows that αT , which for strict adherence to regular behavior should be independent of T , decreases slowly with increasing temperature.

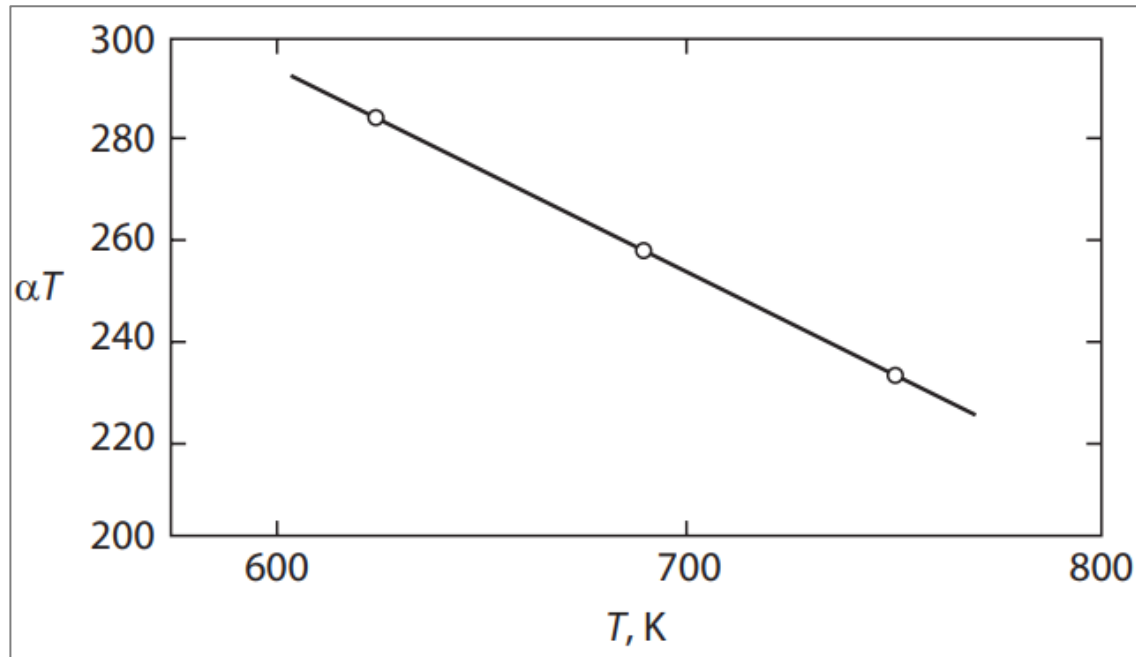


Figure 9.16 The variation of the product αT with T in the system Sn–Tl, showing that the system does not conform to a regular solution, since αT depends on T .

9.9. Regular Solutions

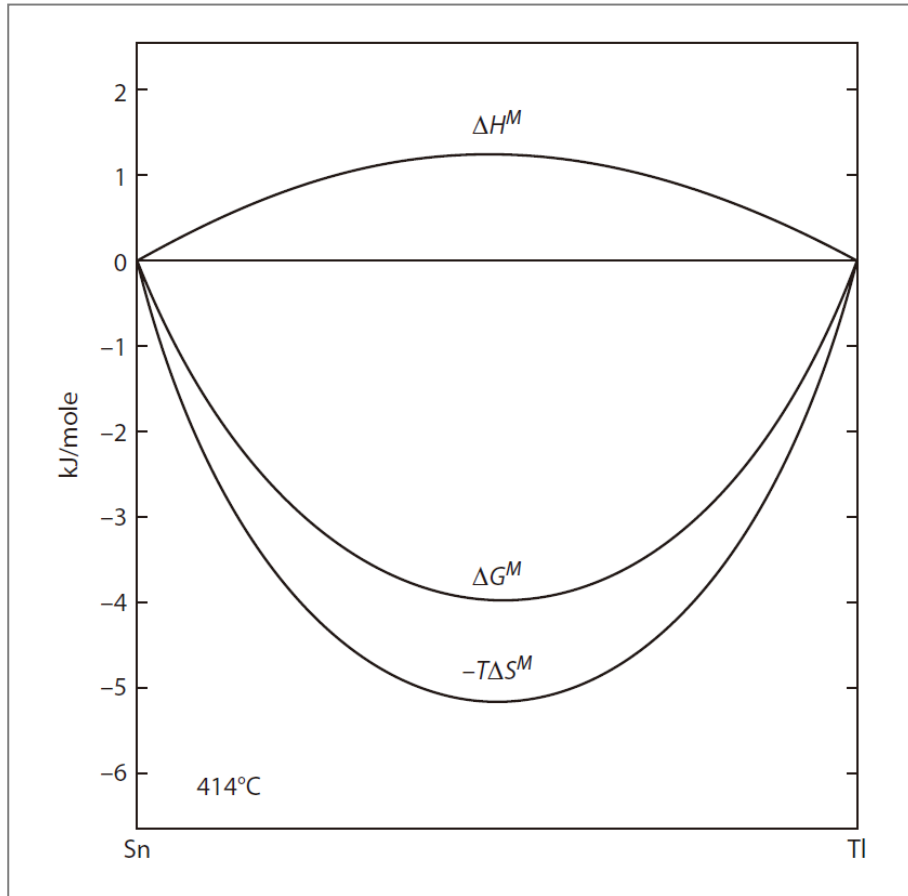


Figure 9.17 shows the variations, with composition, of ΔG^M , ΔH^M , and $-T\Delta S^M$ for the liquid system Sn–Ti at 414 °C. The system exhibits positive deviations from ideality.

9.9. Regular Solutions

- ✓ It is to be noted that a parabolic form for ΔH^M or G^{XS} should not be taken as being a demonstration that the solution is regular, as it is frequently found that ΔH^M or G^{XS} can be adequately expressed by means of the relations

$$\Delta H^M = bX_A X_B \text{ or } G^{XS} = b'X_A X_B$$

where b and b' are unequal, in which case

$$\Delta S^M \neq \Delta S^{M,id}$$

- If **only** configurational entropy is considered, then $\Delta S^M < \Delta S^{M,id}$.
- If, however, other aspects of entropy exist in the alloy (e.g., magnetic spins, defects, volume effects), ΔS^M may be greater than $\Delta S^{M,id}$.

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9.10. A Statistical Model of Solutions

- ✓ Regular solution behavior can be understood by the application of the statistical mixing model, to **two components** which have **equal molar volumes** and which do not exhibit a change in molar volume when mixed; that is, $\Delta \bar{V}_i^M = 0$.
- ✓ In both the pure state and in solution, the **interatomic forces** are **assumed** to exist only between neighboring atoms (near-neighbor model), in which case the energy of the solution is the sum of the interatomic near-neighbor bond energies.

9.10. A Statistical Model of Solutions

- ✓ Consider **1 mole** of a mixed crystal containing N_A atoms of A and N_B atoms of B , such that

$$X_A = \frac{N_A}{N_A + N_B} = \frac{N_A}{N_0} \text{ and } X_B = \frac{N_B}{N_0}$$

where N_0 is Avogadro's number.

The mixed crystal, or solid solution, contains three types of atomic bonds:

1. A – A bonds with energy E_{AA}
2. B – B bonds with energy E_{BB}
3. A – B bonds with energy E_{AB}

9.10. A Statistical Model of Solutions

- ✓ By considering the relative zero of energy to be that when the atoms are infinitely far apart, the bond energies E_{AA} , E_{BB} , and E_{AB} are *negative quantities*.
- ✓ Let the **coordination number** of an atom in the crystal be z ; that is, each atom has z nearest neighbors.
- ✓ If, in the solution, there are P_{AA} A–A bonds, P_{BB} B–B bonds, and P_{AB} A–B bonds, the energy of the solution, E , is obtained as the linear combination
$$E = P_{AA}E_{AA} + P_{BB}E_{BB} + P_{AB}E_{AB}$$
- ✓ The problem of calculating E becomes one of calculating the values of P_{AA} , P_{BB} , and P_{AB} .

9.10. A Statistical Model of Solutions

The number of A atoms $\times$ the number of bonds per atom
= The number of A-B bonds $+ 2 \times$ the number of A-A bonds

(The factor 2 arises because each A-A bond involves two A atoms)



$$zN_A = P_{AB} + 2P_{AA}$$

or

$$P_{AA} = \frac{zN_A}{2} - \frac{P_{AB}}{2}$$

✓ Similarly, for B, $zN_B = P_{AB} + 2P_{BB}$, or

$$P_{BB} = \frac{zN_B}{2} - \frac{P_{AB}}{2}$$

9.10. A Statistical Model of Solutions

$$E = P_{AA}E_{AA} + P_{BB}E_{BB} + P_{AB}E_{AB}$$

$$P_{AA} = \frac{zN_A}{2} - \frac{P_{AB}}{2} \quad \downarrow \quad P_{BB} = \frac{zN_B}{2} - \frac{P_{AB}}{2}$$

$$\begin{aligned} E &= \left(\frac{zN_A}{2} - \frac{P_{AB}}{2} \right) E_{AA} + \left(\frac{zN_B}{2} - \frac{P_{AB}}{2} \right) E_{BB} + P_{AB}E_{AB} \\ &= \frac{1}{2} zN_A E_{AA} + \frac{1}{2} zN_B E_{BB} + P_{AB} \left[E_{AB} - \frac{1}{2} (E_{AA} + E_{BB}) \right] \end{aligned}$$

9.10. A Statistical Model of Solutions

- ✓ Consider now the energies of the **pure components** before mixing.

With N_A atoms in pure A,

$2 \times$ the number of A-A bonds

= the number of atoms $\times$ the number of bonds per atom



$$P_{AA} = \frac{1}{2} z N_A$$

Similarly, for N_B atoms in pure B, $P_{BB} = \frac{1}{2} z N_B$

9.10. A Statistical Model of Solutions

$$\begin{aligned}\Delta E^M &= (\text{the energy of the solution}) - (\text{the energy of the unmixed components}) \\ &= P_{AB} \left[E_{AB} - \frac{1}{2} (E_{AA} + E_{BB}) \right]\end{aligned}$$

✓ For the mixing process,

$$\Delta H^M = \Delta E^M - P\Delta V^M$$

Since it has been stipulated that $\Delta V^M = 0$, then

$$\Delta H^M = \Delta E^M = P_{AB} \left[E_{AB} - \frac{1}{2} (E_{AA} + E_{BB}) \right]$$

9.10. A Statistical Model of Solutions

$$\Delta H^M = \Delta E^M = P_{AB} \left[E_{AB} - \frac{1}{2} (E_{AA} + E_{BB}) \right]$$

- ✓ For given values of E_{AA} , E_{BB} , and E_{AB} , ΔH^M depends on P_{AB} , and further that, for the solution to be ideal (i.e., for $\Delta H^M = 0$),

$$E_{AB} = \frac{1}{2} (E_{AA} + E_{BB})$$

- ✓ Contrary to the preliminary discussion in Section 9.2 which suggested that ideal mixing required the condition $E_{AB} = E_{AA} = E_{BB}$, it is seen that a **sufficient condition** is that E_{AB} be the average of E_{AA} and E_{BB} .

9.10. A Statistical Model of Solutions

$$\Delta H^M = \Delta E^M = P_{AB} \left[E_{AB} - \frac{1}{2} (E_{AA} + E_{BB}) \right]$$

- If $|E_{AB}| > |1/2(E_{AA} + E_{BB})|$, ΔH^M is a **negative** quantity, corresponding to negative deviations from Raoultian ideal behavior,
- If $|E_{AB}| < |1/2(E_{AA} + E_{BB})|$, ΔH^M is a **positive** quantity, corresponding to positive deviations from Raoultian ideality.
- If $\Delta H^M = 0$, then the mixing of the N_A atoms with the N_B atoms of B is **random**. In this case we have

$$\Delta S^M = \Delta S^{M,id} = -R (X_A \ln X_A + X_B \ln X_B)$$

9.10. A Statistical Model of Solutions

- ✓ In solutions which exhibit relatively small deviations from ideal behavior (i.e., $|\Delta H^M| \leq RT$), it can be assumed that the mixing of the atoms is also approximately random, in which case P_{AB} can be calculated as follows.
- ✓ Consider two neighboring lattice sites in the crystal labeled 1 and 2. The probability that site 1 is occupied by an A atom is

$$\frac{\text{The number of A atoms in the crystal}}{\text{The number of lattice sites in the crystal}} = \frac{N_A}{N_0} = X_A$$

- ✓ Similarly, the probability that site 2 is occupied by a B atom is X_B . The probability that site 1 is occupied by an A atom and site 2 is simultaneously occupied by a B atom is thus $X_A X_B$.

9.10. A Statistical Model of Solutions

- ✓ The probability that site 1 is occupied by a B atom and site 2 is simultaneously occupied by an A atom is also $X_A X_B$. Thus, the probability that a neighboring pair of sites contains an $A-B$ pair is $2X_A X_B$.
- ✓ By a similar argument, the probability that the neighboring sites contain an $A-A$ pair is X_A^2 and that the neighboring sites contain a $B-B$ pair is X_B^2 . The probability that the neighboring sites contain an $A-B$ pair or an $A-A$ pair or a $B-B$ pair is

$$\begin{aligned} &X_A^2 + 2X_A X_B + X_B^2 \\ &= (X_A + X_B)^2 \\ &= 1 \end{aligned}$$

9.10. A Statistical Model of Solutions

- ✓ Since a mole of crystal contains $\frac{1}{2} zN_O$ pairs of lattice sites, then

The number of A-B pairs

= The number of pairs of sites $\times$ the probability of an A-B pair

- ✓ That is

$$P_{AB} = \frac{1}{2} zN_O \times 2X_A X_B = zN_O X_A X_B$$

- ✓ Similarly,

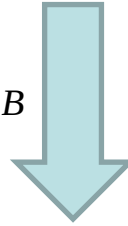
$$P_{AA} = \frac{1}{2} zN_O \times X_A^2 = \frac{1}{2} zN_O X_A^2$$

$$P_{BB} = \frac{1}{2} zN_O X_B^2$$

9.10. A Statistical Model of Solutions

$$\Delta H^M = \Delta E^M = P_{AB} \left[E_{AB} - \frac{1}{2}(E_{AA} + E_{BB}) \right]$$

$$P_{AB} = \frac{1}{2} z N_O \times 2 X_A X_B = z N_O X_A X_B$$



$$\Delta H^M = z N_O X_A X_B \left[E_{AB} - \frac{1}{2}(E_{AA} + E_{BB}) \right]$$

If we set W_1 as

$$W_1 = \left[E_{AB} - \frac{1}{2}(E_{AA} + E_{BB}) \right]$$



$$\Delta H^M = z N_O W_1 X_A X_B$$

9.10. A Statistical Model of Solutions

$$\Delta H^M = \underline{zN_0W_1} X_A X_B$$

- ✓ This shows that ΔH_M is a parabolic function of composition. Since random mixing is assumed, the statistical model corresponds to the regular solution model; that is,

$$\Delta H^M = G^{XS} = \underline{zN_0W_1} X_A X_B$$

$$\Delta H^M = \alpha X_A X_B$$



$$\alpha = zN_0W_1$$

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

- ✓ As mentioned previously, a major flaw in the regular solution model is that, in counting neighbors to obtain the enthalpy of mixing as well as the entropy of mixing, it is assumed that the atoms are randomly placed on the lattice.
- ✓ But if the expression
$$E_{AB} - \frac{1}{2}(E_{AA} + E_{BB})$$
 is not zero, the atoms will not be random!
- ✓ For atomic ordering transformations, this can be improved by including a thermodynamic variable which accounts for the nonrandom configuration of the atoms. This factor is the long-range order parameter (LRO), η .

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

- ✓ Consider the ordering transformation of a random $A-B$ solution phase (fully disordered) with a BCC (A2) structure to the fully ordered phase with the CsCl (B2) structure

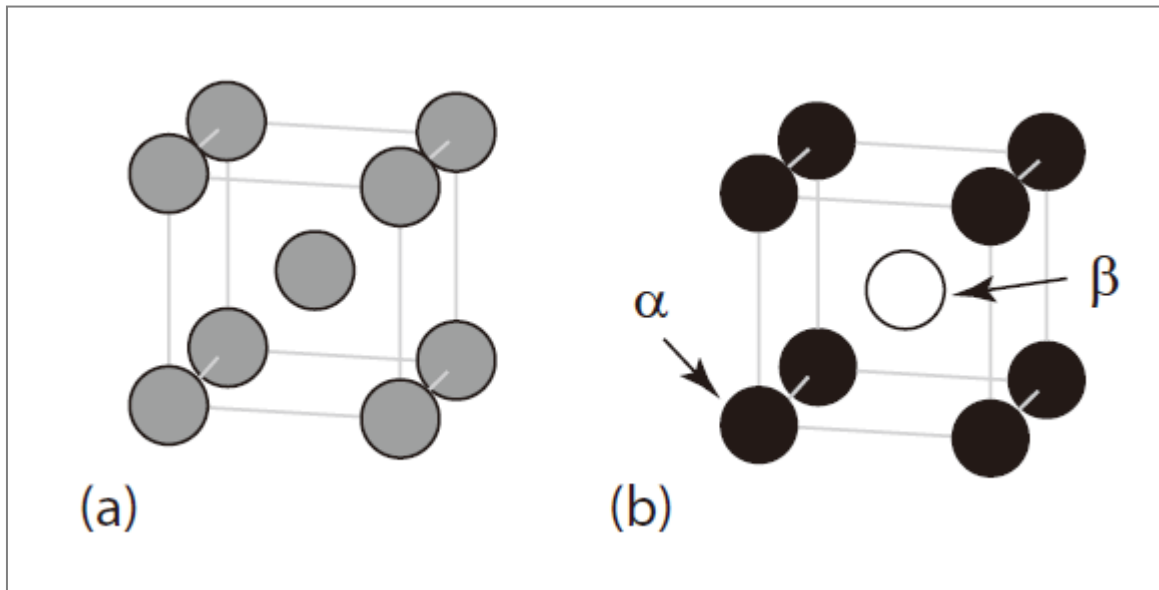


Figure 9.18 (a) BCC (A2) and (b) CsCl (B2) structures. The white and black atoms are randomly arranged in the BCC structure, but in the CsCl structure, the black atoms (A atoms) sit on the α sites and the white atoms (B atoms) are on the β sites.

- ✓ The LRO parameter η is defined to be

$$\eta = \frac{r_{\alpha} - X_A}{Y_{\beta}} = \frac{r_{\beta} - X_B}{Y_{\alpha}}$$

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

$$\eta = \frac{r_\alpha - X_A}{Y_\beta} = \frac{r_\beta - X_B}{Y_\alpha}$$

- ✓ Here r_α (r_β) is the fraction of α (β) sites that are “rightly” occupied, X_i is the fraction of the i th component, and Y_α (Y_β) is the fraction of α (β) sites in the lattice.
- ✓ For the BCC→B2 transformation, Y_α and Y_β are $1/2$. Thus, for an equiatomic alloy ($X_A=X_B=1/2$), we have

$$\eta = 2r_\alpha - 1 = 2r_\beta - 1$$

- If all the α sites are occupied by A atoms, $r_\alpha=1$ and $\eta=1$.
- If all the α sites are occupied by B atoms, $r_\alpha=0$ and $\eta=-1$.
- If the α sites are randomly occupied, $r_\alpha=1/2$ and $\eta = 0$.

The case for $\eta = -1$ corresponds to a translational domain of the B2-ordered system.

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

- ✓ The expression for the enthalpy of formation of a system from the pure A and pure B components was given previously as

$$\Delta H^M = zN_0 X_A X_B W_1$$

where $W_1 = E_{AB} - \frac{1}{2}(E_{AA} + E_{BB})$. For $W_1 < 0$, atomic ordering is favored. For $W_1 = 0$, the alloy is random, and for $W_1 > 0$, clustering is favored.

- ✓ If we include the effect of atomic ordering on the occupation of the sites, we obtain

$$\Delta H^M = zN_0 W_1 \left(X_A X_B + \frac{\eta^2}{4} \right)$$

- ✓ For a full order of an equiatomic alloy, this yields $\Delta H^M = zN_0 W_1 / 2$. This can be seen to be twice as negative as the expression which does not include the order parameter.

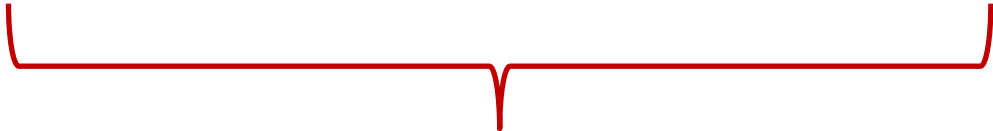
9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

- ✓ The effect of ordering also can be included in the entropy of mixing as follows:

$$\Delta S^M(\eta) = -\frac{Nk_B}{2} \left\{ \left(X_A + \frac{\eta}{2} \right) \ln \left(X_A + \frac{\eta}{2} \right) + \left(X_B - \frac{\eta}{2} \right) \ln \left(X_B - \frac{\eta}{2} \right) \right. \\ \left. + \left(X_A - \frac{\eta}{2} \right) \ln \left(X_A - \frac{\eta}{2} \right) + \left(X_B + \frac{\eta}{2} \right) \ln \left(X_B + \frac{\eta}{2} \right) \right\}$$

- ✓ For full order of an equiatomic alloy this yields $\Delta S_{conf}=0$, since the pure elements have zero configurational entropy and the fully ordered B2 phase has zero configurational entropy.

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

$$\Delta H^M = zN_0W_1\left(X_A X_B + \frac{\eta^2}{4}\right) \quad \Delta S^M(\eta) = -\frac{Nk_B}{2}\left\{\left(X_A + \frac{\eta}{2}\right)\ln\left(X_A + \frac{\eta}{2}\right) + \left(X_B - \frac{\eta}{2}\right)\ln\left(X_B - \frac{\eta}{2}\right) + \left(X_A - \frac{\eta}{2}\right)\ln\left(X_A - \frac{\eta}{2}\right) + \left(X_B + \frac{\eta}{2}\right)\ln\left(X_B + \frac{\eta}{2}\right)\right\}$$


$$\Delta G^M(\eta, T) = zN_0W_1\left(X_A X_B + \frac{\eta^2}{4}\right) + \frac{Nk_B T}{2}\left\{\left(X_A + \frac{\eta}{2}\right)\ln\left(X_A + \frac{\eta}{2}\right) + \left(X_B - \frac{\eta}{2}\right)\ln\left(X_B - \frac{\eta}{2}\right) + \left(X_A - \frac{\eta}{2}\right)\ln\left(X_A - \frac{\eta}{2}\right) + \left(X_B + \frac{\eta}{2}\right)\ln\left(X_B + \frac{\eta}{2}\right)\right\}$$

- ✓ Since ΔG^M is a function of the order parameter, we take its derivative with respect to it and set it equal to zero:

$$\frac{\partial \Delta G^M}{\partial \eta} = \frac{zN_0W_1\eta}{2} + \frac{N_0k_B T}{2} \ln\left(\frac{1+\eta}{1-\eta}\right) = 0$$

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

- ✓ As η approaches zero, the temperature approaches the transition temperature. For small η ,

$$\ln\left(\frac{1+\eta}{1-\eta}\right) = 2\left[\eta + \frac{\eta^3}{3} + \dots\right] \approx 2\eta$$

$$\frac{\partial \Delta G^M}{\partial \eta} = \frac{zN_0W_1\eta}{2} + \frac{N_0k_BT}{2} \ln\left(\frac{1+\eta}{1-\eta}\right) = 0$$



$$\frac{zN_0W_1\eta}{2} + \frac{2N_0k_BT\eta}{2} = 0$$



$$T_C = -\frac{4W_1}{k_B}$$

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

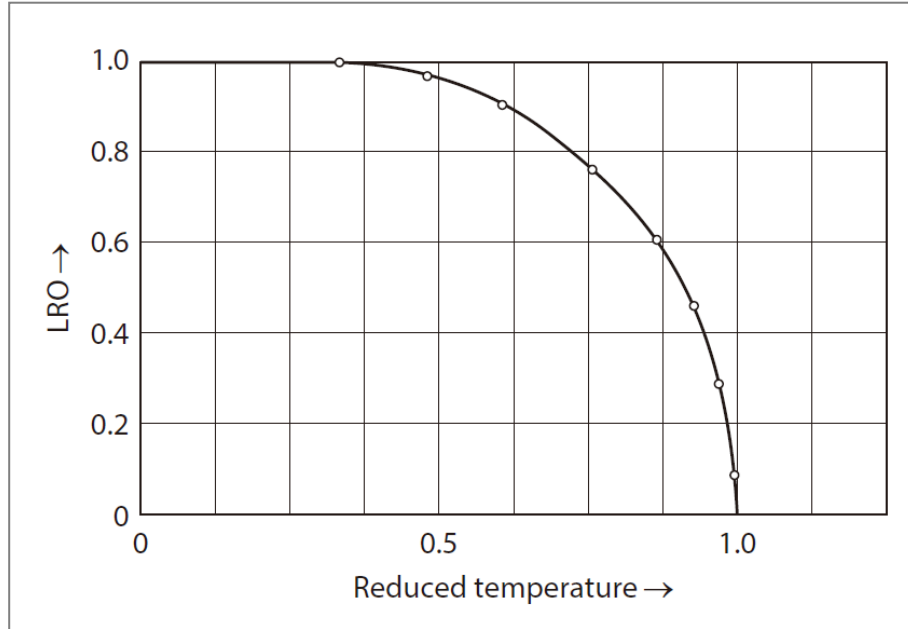


Figure 9.19 The atomic LRO parameter η vs. temperature plot for a Bragg–Williams model. Note the infinite slope as T approaches T_C and the zero slope as T approaches 0 K.

- ✓ Thus, this model produces an atomic ordering phase transition at T_C .
- ✓ Above this temperature, the equilibrium phase is disordered BCC.
- ✓ Below this temperature, the equilibrium phase is ordered CsCl with an LRO parameter η that increases continuously with decreasing temperature and which approaches unity as T approaches 0 K.

$$T_C = -\frac{4W_1}{k_B}$$

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

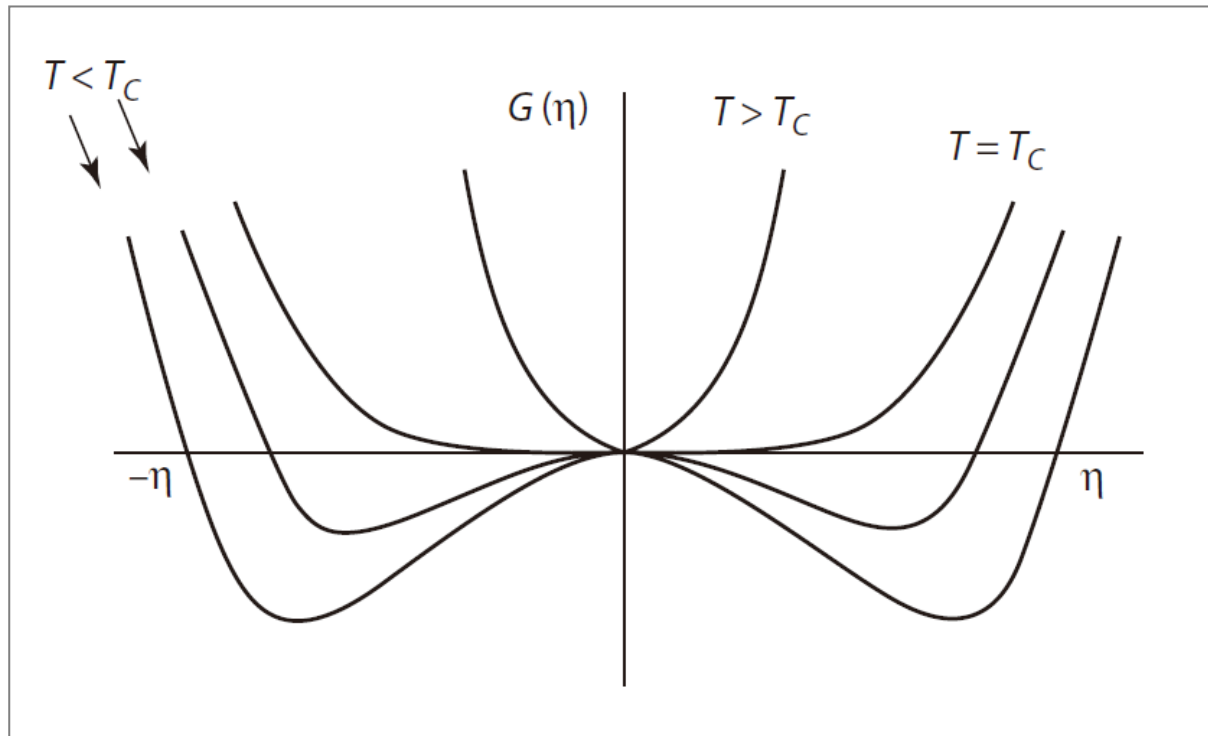


Figure 9.20 Gibbs free energy of mixing, ΔG_M , vs. order parameter at various temperatures relative to the critical temperature.

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

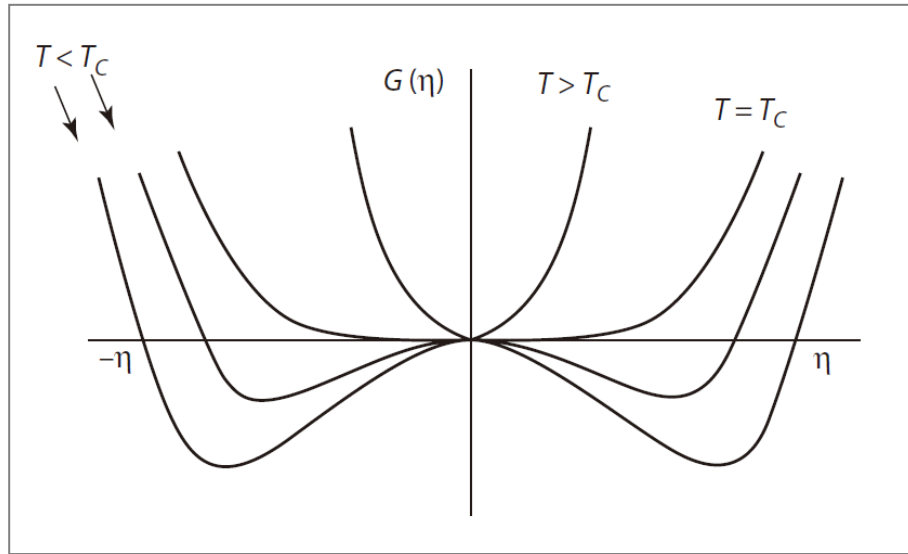


Figure 9.20 Gibbs free energy of mixing, ΔG_M , vs. order parameter at various temperatures relative to the critical temperature.

- ✓ At temperatures above T_C , the minimum in ΔG^M occurs at $\eta = 0$; that is, the disordered phase is the stable one.
- ✓ Below T_C , two minima appear at $\pm \eta_{eq}$.
- ✓ A maximum appears at $\eta = 0$, showing that the disordered phase is unstable below T_C .
- ✓ As the temperature is lowered, η_{eq} increases until at $T = 0$ K, $\eta_{eq} = 1$.

9.10.1. Extensions of the Regular Solution Model: the Atomic Order Parameter

- ✓ we notice that ΔG^M versus η can be written in a Taylor series form as follows:

$$\Delta G^M = a + b\eta^2 + c\eta^4$$

where a is a constant of integration, b is a function of T , and c can be considered a positive constant.

9.10.2. Including Second-Neighbor Interactions

- ✓ Another extension of this solution model can be made by adding second-neighbor interactions. Previously, we defined first-neighbor interactions as

$$W_1 = E_{AB}^{(1)} - \frac{1}{2} \left(E_{AA}^{(1)} + E_{BB}^{(1)} \right)$$

- ✓ Second-neighbor interaction energies can be written as

$$W_2 = E_{AB}^{(2)} - \frac{1}{2} \left(E_{AA}^{(2)} + E_{BB}^{(2)} \right)$$

- ✓ This expression can be added to the one for the heat of transformation used previously to obtain

$$\Delta H^M = N_0 X_A X_B \left[z_1 W_1 + z_2 W_2 \right] + \frac{N_0 \eta^2}{2} \left[z_1 W_1 - z_2 W_2 \right]$$

9.10.2. Including Second-Neighbor Interactions

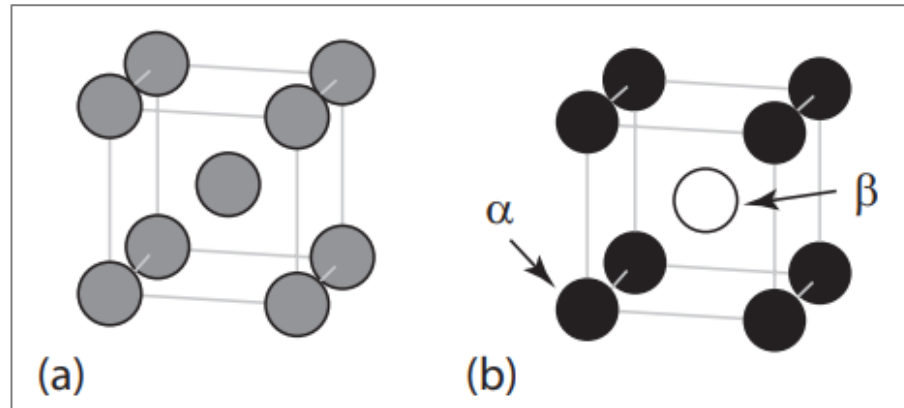


Figure 9.18 (a) BCC (A2) and (b) CsCl (B2) structures. The white and black atoms are randomly arranged in the BCC structure, but in the CsCl structure, the black atoms (*A* atoms) sit on the α sites and the white atoms (*B* atoms) are on the β sites.

- ✓ It can be seen that for the B2-ordered structure, although $W_1 < 0$ (favoring opposite first neighbors), $W_2 > 0$ (favoring the same second neighbors). The details and further work on these expressions can be found in Soffa and Laughlin and Soffa, Laughlin, and Singh,* where the authors analyze the transformations $\text{FCC} \rightarrow \text{L1}_2$ and $\text{FCC} \rightarrow \text{L1}_0$.

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9.11. Subregular Solutions

$$\Delta H^M = \alpha X_A X_B$$

$$\Delta H^M = ZN_O W_1 X_A X_B$$

- ✓ In the **regular solution** model, the **constant value** of α , gives a parabolic variation of ΔH^M , and the ideal entropy of mixing lead to variations of G^{XS} and ΔG^M which are **symmetrical** about the composition $X_A = 0.5$.
- ✓ The model can be made more flexible by arbitrarily allowing α to vary with composition such as

$$\underline{\alpha = a + bX_B + cX_B^2 + dX_B^3 + \dots}$$

- ✓ The so-called subregular solution model is one in which the values of all of the constants, other than a and b , are zero.

9.11. Subregular Solutions

- ✓ Thus, the **subregular solution** model gives the molar excess Gibbs free energy of formation of a binary **A–B** solution as

$$G^{XS} = (a + bX_B) X_A X_B$$

- ✓ This Equation is an empirical equation; that is, the constants **a** and **b** have no physical significance and are simply parameters.

$$G^{XS} = (a + bX_B) X_A X_B$$
$$\bar{G}_A = G - X_B \frac{dG}{dX_B} \quad \downarrow \quad \bar{G}_B = G + X_A \frac{dG}{dX_B}$$

The partial molar excess Gibbs free energies of the components A and B as

$$\bar{G}_A^{XS} = aX_B^2 + bX_B^2(X_B - X_A) \quad \bar{G}_B^{XS} = aX_A^2 + 2bX_A^2X_B$$

9.11. Subregular Solutions

$$G^{XS} = (a + bX_B) X_A X_B$$

✓ The maxima and/or minima in the curves occur at

$$\frac{dG^{XS}}{dX_B} = 0$$

$$G^{XS} = aX_B + (b - a)X_B^2 - bX_B^3$$



$$\frac{dG^{XS}}{dX_B} = a + 2(b - a)X_B - 3bX_B^2 = 0$$



$$X_B = \frac{2(b - a) \pm 2\sqrt{b^2 + ab + a^2}}{6b}$$

9.11. Subregular Solutions

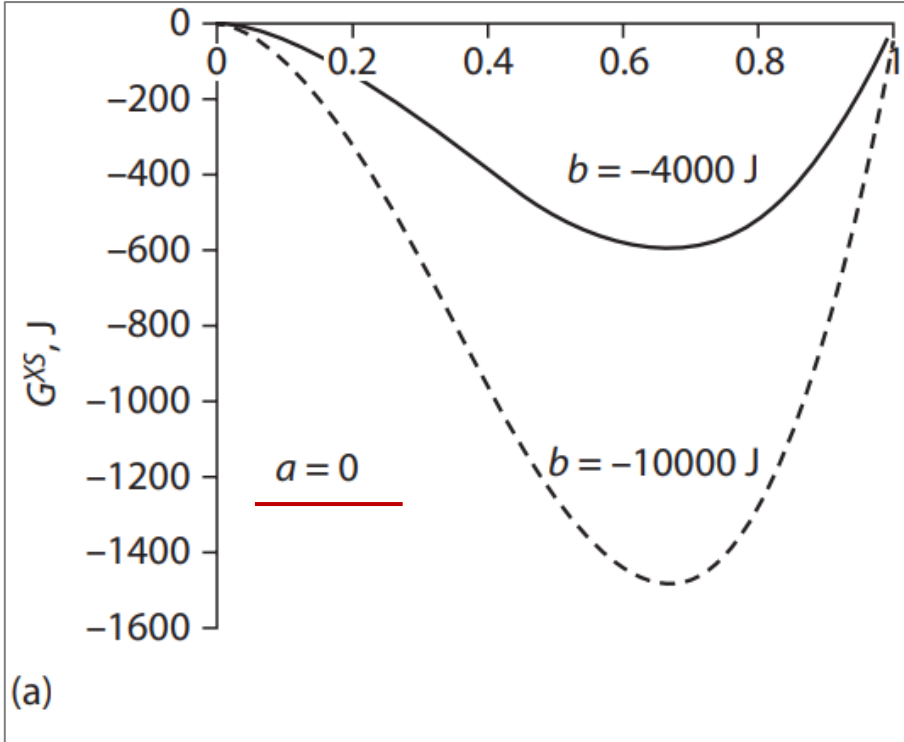


Figure 9.21 Excess molar Gibbs free energy curves generated by the subregular solution model for various values of a and b .

- ✓ With $a = 0$ and $b \neq 0$, the minimum in the curve occurs at $X_B = 2/3$, if $b \neq 0$.

9.11. Subregular Solutions

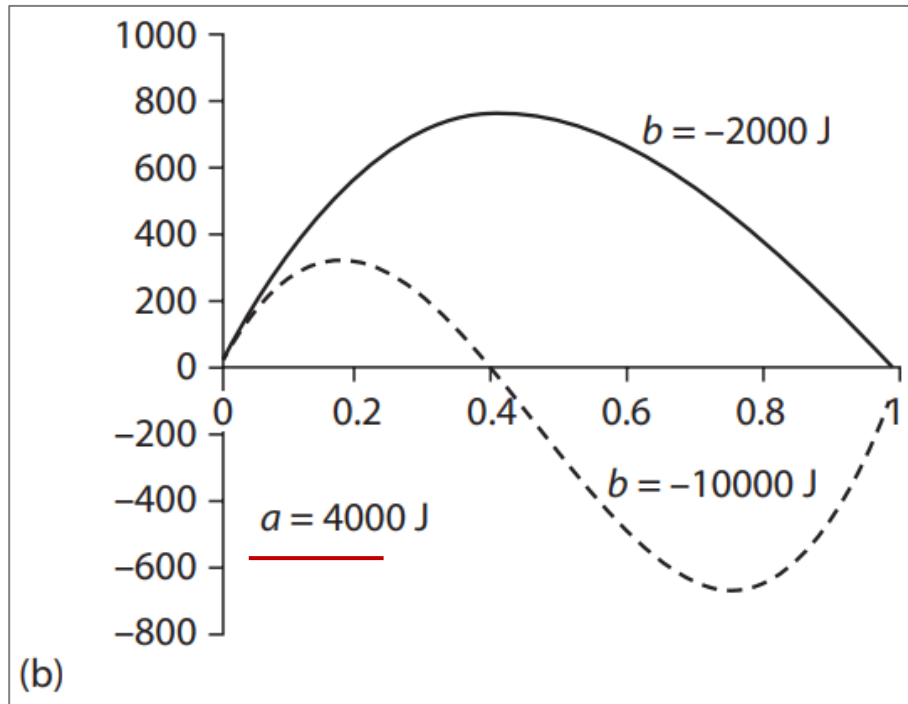


Figure 9.21 Excess molar Gibbs free energy curves generated by the subregular solution model for various values of a and b .

- ✓ With $a = 4000$ J and $b = -10,000$ J, a maximum occurs in the curve at $X_B = 0.17$ and a minimum occurs at $X_B = 0.76$.
- ✓ If, however, $b = -2,000$ J, no minimum occurs, only a maximum.

9.11. Subregular Solutions

- ✓ The influence of **temperature** on the behavior of subregular solutions is accommodated by introducing a third constant, τ , to give the molar excess Gibbs free energy of mixing as

$$G^{XS} = (a_0 + b_0 X_B) X_A X_B \left(1 - \frac{T}{\tau} \right)$$

- ✓ The molar **excess entropy** of mixing is thus

$$S^{XS} = - \frac{\partial G^{XS}}{\partial T} = \frac{(a_0 + b_0) X_A X_B}{\tau}$$

- ✓ The molar enthalpy of mixing (which is also the molar excess enthalpy of mixing) is given by

$$\Delta H^M = G^{XS} + TS^{XS} = (a_0 + b_0 X_B) X_A X_B \left(2 - \frac{T}{\tau} \right)$$

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9.12. Modified Regular Solution Model for Application to Polymers

◆ 9.12.1. the Flory–Huggins Model*

- ✓ In 1942, Flory and Huggins independently modified the regular solution model to account for some of the differences that polymers present to the modeling of the thermodynamics of the mixing of polymers.
- ✓ The main difference lies in the entropy of mixing term, which in modern notation is written as

$$\Delta S^{mix} = -R \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B \right)$$

- ✓ where ϕ_i is the volume fraction of the i th monomers which make up the polymer and N_i is the average number of *monomers* in the length of the molecule (the degree of polymerization).

9.12. Modified Regular Solution Model for Application to Polymers

- ✓ For polymers, the entity which is randomly located in the solution is not a single atom or molecule but rather a long chain consisting of the many *-mers* of the polymer.
- ✓ This lowers the number of ways that the polymer can be arranged in a given volume, and it is accounted for by the N_i terms in the expression for the entropy of mixing. This greatly lowers the role of entropic mixing in large polymers.
- ✓ The enthalpy of mixing term was written as $\Delta H^{mix} = \phi_A \phi_A \chi$ where χ accounts for the increase (decrease) in the enthalpy of the solution due to interactions between the polymers.

9.12. Modified Regular Solution Model for Application to Polymers

$$\Delta H^{mix} = \phi_A \phi_B \chi$$

- ✓ This term, combined with the entropy of mixing term, gave rise to the Gibbs free energy of mixing term:*

$$\Delta G^{mix} = RT \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B + \phi_A \phi_B \chi \right)$$

- ✓ It can be seen for polymers of long length (N_i large), the entropy of mixing term is not as large as that for solutions of single atoms or small molecules.
- ✓ Since most polymers have positive enthalpies of mixing, this accounts for the limited miscibility of many polymers.

9.12. Modified Regular Solution Model for Application to Polymers

$$\Delta H^{mix} = \phi_A \phi_B \chi$$

- ✓ This term, combined with the entropy of mixing term, gave rise to the Gibbs free energy of mixing term:*

$$\Delta G^{mix} = RT \left(\frac{\phi_A}{N_A} \ln \phi_A + \frac{\phi_B}{N_B} \ln \phi_B + \phi_A \phi_B \chi \right)$$

- ✓ The expression shows that if one of the polymers is not very long, it can act as the solvent and the long-chained polymer as the solute (a polymer blend).
- ✓ This enables much more miscibility for the polymer solutions, as the entropy term becomes more significant to the polymer thermodynamics.

Homework

1. The molar excess Gibbs free energy of formation of solid solutions in the system Au–Ni can be represented by

$$G^{XS} = X_{Ni} X_{Au} (24140 X_{Au} + 38280 X_{Ni} - 14230 X_{Au} X_{Ni}) \left(1 - \frac{T}{2660} \right) \text{ J}$$

Calculate the activities of Au and Ni in the alloy of $X_{Au} = 0.5$ at 1100 K.

2. It has been found that in a certain solution, the activity $a_A = X_A$ over a certain range of composition. Determine the relationship between a_B and X_B over the same range of composition

Homework

3. At a certain temperature, T , the A–B system exhibits regular solution behavior. The activity coefficient of A is given by

$$\ln \gamma_A = -b(1 - X_A)^2$$

where b is a constant at the given T .

Compute the corresponding equation for the variation of γ_B with composition at the same temperature. Be sure to state the justification for the steps of your solution.

Thermodynamics of Materials & Phase transformations



The End

9.13. Summary

1. **Raoult's law** is $p_i = X_i p_i^\circ$ and a component of a solution that conforms with this law is said to exhibit Raoultian behavior. In all solutions, the behavior of the component i approaches Raoult's law as $X_i \rightarrow 1$.
2. **Henry's law** is $p_i = k'X_i$, and a component of a solution which conforms with this equation is said to exhibit Henrian behavior. In all solutions, the behavior of the component i approaches Henry's law as $X_i \rightarrow 0$. In a binary solution, Henry's law is obeyed by the solute in that composition range over which Raoult's law is obeyed by the solvent.

9.13. Summary

3. The activity of the component i in a solution, with respect to a given standard state, is the ratio of the vapor pressure (strictly, the fugacity) of i exerted by the solution to the vapor pressure (the fugacity) of i in the given standard state. If the standard state is chosen as being pure i , then $a_i = p_i/p_i^\circ$. An activity is thus a ratio, and its introduction effects a normalization of the vapor pressure exerted by the component i in the solution. In terms of activity, Raoult's law is $a_i = X_i$, and Henry's law is $a_i = k^B X_i$.

9.13. Summary

4. The difference between the value of an extensive thermodynamic property per mole of i in a solution and the value of the property per mole of i in its standard state is called the partial molar property change of i for the solution process; that is, if Q is any extensive thermodynamic property, the change in the property due to the solution of 1 mole of i is $\overline{\Delta Q}_i^M = \bar{Q}_i - Q_i^\circ$. In the case of the Gibbs free energy, $\overline{\Delta G}_i^M = \bar{G}_i - G_i^\circ$. This difference in the molar Gibbs free energy is related to the activity of i in solution, with respect to the standard state, as $\overline{\Delta G}_i^M = RT \ln a_i$, and $\overline{\Delta G}_i^M$ is called the partial molar Gibbs free energy of solution of i .

9.13. Summary

4. The change in the Gibbs free energy accompanying the formation of 1 mole of solution from the pure components i (called the integral Gibbs free energy change) is $\Delta G^M = \sum_i X_i \Delta \bar{G}_i^M$, so that for the binary A–B, $\Delta G^M = X_A \Delta \bar{G}_A^M + X_B \Delta \bar{G}_B^M$. Since $\Delta \bar{G}_A^M = RT \ln a_A$, then

$$\Delta G^M = RT (X_A \ln X_A + X_B \ln X_B)$$

- ✓ In a Raoultian solution, since $a_i = X_i$, then

$$\Delta G^M = RT (X_A \ln X_A + X_B \ln X_B)$$

- ✓ For any general extensive thermodynamic property,

$$Q, \Delta Q^M = \sum_i X_i \Delta \bar{Q}_i^M$$

9.13. Summary

5. An ideal solution has the following properties:

$$\square a_i = X_i$$

$\square V_i = V_i^0$ (i.e., there is no change in volume when the components are mixed)

$$\square \overline{H}_i^M = H_i^0 \text{ (i.e., there is zero heat of mixing)}$$

$$\square \Delta G^{M,id} = RT (X_A + X_B \ln X_B)$$

✓ Since $\Delta S^{M,id} = -(\partial \Delta G^{M,id} / \partial T)$, $\partial S^{M,id} = -R \ln \sum_i X_i$, in an ideal solution, $\Delta \overline{S}_i^M = -R \ln X_i$. $\Delta S^{M,id}$ is thus independent of temperature and is simply an expression for the maximum number of spatial configurations available to the system.

9.13. Summary

6. The thermodynamic behavior of non-Raoultian solutions is dealt with by introducing the activity coefficient, γ , which for the component i is defined as $\gamma_i = a_i/X_i$. The coefficient γ_i , which can have values of greater or less than unity, thus quantifies the deviation of i from Raoultian behavior. Since $\ln a_i = \ln X_i + \ln \gamma_i$,

$$\frac{d \ln a_i}{d\left(\frac{1}{T}\right)} = \frac{\Delta \overline{H}_i^M}{R} = \frac{d \ln \gamma_i}{d\left(\frac{1}{T}\right)}$$

9.13. Summary

- ✓ Thus, if dy_i/dT is positive, $\Delta\overline{H}_i^M$ is negative, and if dy_i/dT is negative, $\Delta\overline{H}_i^M$ is positive. The magnitude of the heat of formation of a nonideal solution is determined by the magnitudes of the deviations of the components of the solution from ideal behavior. Nonideal components approach Raoultian behavior with increasing temperature. Thus, if $\gamma_i < 1$, then dy_i/dT is positive, and if $\gamma_i > 1$, dy_i/dT is negative. Solutions, the components of which exhibit negative deviations from Raoult's law, form exothermically; that is, $\Delta H^M < 0$, and those that exhibit positive deviations form endothermically.

9.13. Summary

7. The Gibbs–Duhem relationship is $\sum_i X_i d\bar{Q}_i = 0$ at constant temperature and pressure, where $\bar{Q}_i$ is the partial molar value of the extensive thermodynamic function Q of the solution component i . The excess value of an extensive thermodynamic property of a solution is the difference between the actual value and the value that the property would have if the components obeyed Raoult's law. Thus, for the general function Q , $Q^{XS} = Q - Q^{id}$, or for the Gibbs free energy, $G^{XS} = G - G^{id}$, or $G^{XS} = \Delta G^M - \Delta G^{M,id}$. Since $\gamma_i = a_i/X_i$, then

$$G^{XS} = RT \sum X_i \ln \gamma_i$$

9.13. Summary

8. A regular solution is one which has an ideal entropy of formation and a nonzero heat of formation from its pure components. The activity coefficients of the components of a regular binary solution are given by the expression

$$RT \ln \gamma_i = \alpha (1 - X_i)^2$$

where α is a temperature-independent constant, the value of which is characteristic of the particular solution.

- ✓ Thus, $\ln \gamma_i$ varies inversely with temperature, and, since $\bar{G}_i^{XS} = \ln \gamma_i$, then $\bar{G}_i^{XS} = \Delta \bar{H}_i^M$, is independent of temperature. Furthermore, the heat of formation of a regular solution, being equal to G^{XS} , is a parabolic function of composition, given by

$$\Delta H^M = G^{XS} = \alpha X_A X_B$$

9.13. Summary

9. Regular solution behavior is predicted by a statistical solution model in which it is assumed that the atoms mix randomly and that the energy of the solution is the sum of the individual interatomic bond energies in the solution. Random mixing can be assumed only if, in the system $A-B$, the $A-B$ bond energy is not significantly different from the average of the $A-A$ and $B-B$ bond energies in the pure components. For any such deviation, the validity of the assumption of random mixing increases with increasing temperature. The statistical model predicts tendency toward Raoultian behavior and Henrian behavior as, respectively, $X_i \rightarrow 1$ and $X_i \rightarrow 0$

9.13. Summary

10. The statistical solution model can be improved by adding the effect of second nearest neighbors. In the case of solutions with ordering tendency (negative deviations from Raoultian behavior), the model for both the molar enthalpy and molar entropy of formation can be improved by adding an LRO parameter to the molar enthalpy and molar entropy terms.

9.13. Summary

11. The subregular solution model is one in which the value of α is assumed to be a linear function of composition, being given by $\alpha = a + bX_B$. The variation of the molar excess Gibbs free energy of mixing is thus given by $G^{XS} = (a + bX_B)X_A X_B$. This can give rise to asymmetric dependence of the molar properties of formation. The constants a and b are curve-fitting parameters and have no physical significance.